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Safe passage of TBM under new NEL tunnels, using a foam grout backfilled advance tunnel

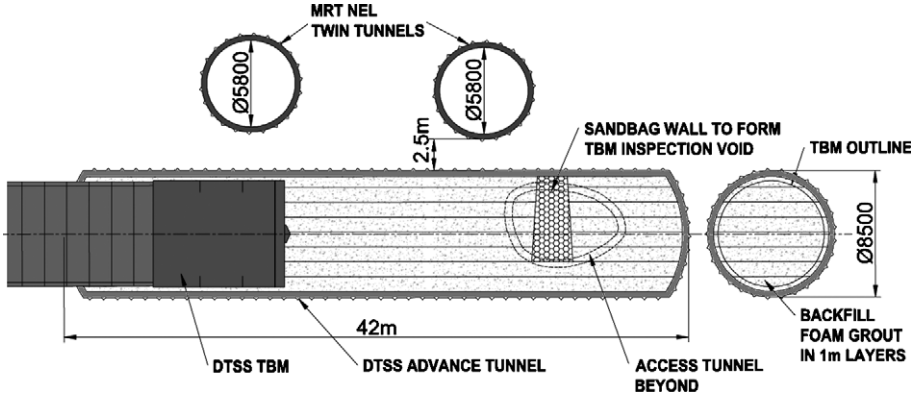
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Contract T-03 of Singapore's Deep Tunnel Sewerage System (DTSS) was one of six DTSS design and build contracts for the Public Utility Board (PUB), of approximate value S$70M, consisting of 5.2 km long × 6.5 m ID × 30 m deep tunnel with six shafts.

One of the challenges on contract T-03 was that the alignment of the 7.2 m diameter bored tunnel crossed beneath the alignment of the newly built MRT North East Line (NEL) 5.8 m diameter twin tunnels, which were undergoing testing for the running of passenger trains during the time of construction of contract T-03. The invert level of the NEL tunnels was 2.5 m above the crown of the T-03 tunnel, therefore precautions were necessary to limit movement of the NEL tunnels during excavation of the DTSS tunnel below them. For this reason the DTSS Advance Tunnel was constructed prior to driving both the MRT NEL and DTSS TBM's, to avoid active tunnelling beneath the NEL tunnels.

It was envisaged that the DTSS contractor would, upon reaching the Advance Tunnel with the TBM, break-in through the endwall of the chamber, "walk" the TBM through the empty chamber and then break out through the farthest endwall to continue tunnelling. This concept requires the fabrication of a steel cradle within the chamber to accept the TBM and enable it to be 'walked' through, the underground erection of a shove frame to re-launch the TBM through the endwall, followed in this case by the later erection of tunnel lining in free air and the backfilling of the large void between the outside of the lining and the walls of the chamber. Additionally the Contractor would also have to



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take into consideration the risk of ground entering the chamber during the TBM break-in and out of the chamber and take appropriate measures to mitigate this risk.

Faced with these challenges, the Contractor proposed to backfill the chamber prior to the TBM arriving, thus eliminating any risk of ground loss into the chamber as the chamber would be full of material. The TBM could travel through the Advance Tunnel unaided, without productivity being seriously affected, and minimising the duration that the TBM was stopped in the zone of influence of the NEL twin tunnels. Foamed grout was selected as the most suitable backfilling material. 1800 m^3 of backfill was required, obtained by the use of 1300 m^3 of grout expanded by 30% with the addition of air. The foamed grout was designed with sufficient strength to support the TBM, but weak enough in shear to avoid unnecessary wear to the cutterhead.

The backfilling operation was completed in 5 weeks, prior to the arrival of the TBM. The TBM successfully passed through the backfilled advance tunnel. There were substantial programme benefits, at a minimum cost, whilst achieving the main objective of not impacting the NEL tunnels above.

Keywords: Advance tunnel; Foamed grout; Backfilling